PROJECT DESIGN — MILESTONE 2

AI-Based Dual Cleanliness Monitoring System

Solid Waste Detection + Liquid Waste Segmentation

1. Proposed Work

The proposed system is an AI-based cleanliness monitoring solution that handles two visually different problems: (1) solid waste such as bottles, plastic, paper, cans and garbage, and (2) visible liquid waste/spills. Instead of using one model for both tasks, the project uses a dual computer-vision pipeline. Solid waste is treated as an object-detection problem, while liquid waste is treated as a segmentation problem because the spill boundary and affected area are important.

2. Objectives

1. Detect and classify common solid-waste objects from images or CCTV frames.
2. Detect and segment visible liquid/spill regions and estimate their affected area.
3. Convert AI outputs into measurable waste quantity and severity information.
4. Generate an interpretable cleanliness score and cleaning status.
5. Support video monitoring and, in the second phase, tracking of persistent waste.
6. Provide visual alerts and analytics without developing a new training database from scratch.

3. Final Model Selection

Module	Selected model	Why it is used
Solid waste	YOLO11	Fast one-stage detection; gives class, bounding box and confidence for multiple waste objects.
Liquid waste	YOLO11-Seg	Pixel-level segmentation is more suitable for irregular liquid/spill regions than a rectangular box.
Video tracking	ByteTrack (Phase 2)	Keeps an ID for the same detected solid object across frames and reduces repeated counting.
Benchmark	U-Net (optional)	Classic segmentation baseline for comparison if time permits.

Model size recommendation: start with YOLO11s for both branches so the complete system can be trained and tested within student-level compute limits. If validation results or available GPU resources justify it, compare YOLO11m. The goal is not simply to use the largest model; the goal is to obtain a strong accuracy–speed trade-off.

4. Public Datasets — No Custom Database Required

The training datasets will be obtained from existing public repositories. The team does not need to build a database from scratch. The datasets can be downloaded, prepared into the required format, and used directly or after class mapping. For the final model, the most important requirement is that the train/validation/test data are kept separate.

Branch	Recommended existing dataset	Size / format	Reason
Solid	Garbage Detection – 6 Waste Categories (Kaggle)	10,464 labeled images; YOLO format	Large, directly usable with YOLO; 6 waste categories; train/valid/test already provided.
Liquid	Spill Instance Segmentation Dataset (Roboflow)	4,500 images; instance segmentation; CC BY 4.0	Large and directly focused on spill detection; classes include water, trash and spill.
Liquid research benchmark	LQDS	5,000 real-world images; pixel masks; 14 liquid classes	Broader liquid segmentation research dataset; useful if

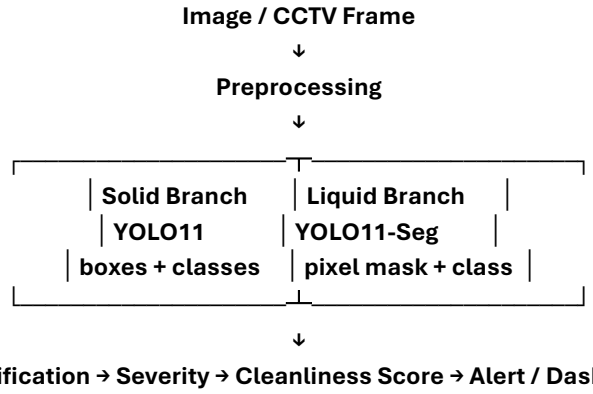
			accessible to the team.

The solid Kaggle dataset contains BIODEGRADABLE, CARDBOARD, GLASS, METAL, PAPER and PLASTIC, with YOLO-format labels and 7,324/2,098/1,042 train/validation/test images. The Roboflow spill dataset contains 4,500 images and three instance-segmentation classes: water, trash and spill. LQDS contains 5,000 real-world images with pixel-level masks across 14 liquid classes and is useful as a research-grade benchmark.

5. Methodology of Proposed Work

The system will be developed in stages. The two AI branches are trained independently first, then their outputs are combined into one cleanliness-analysis module.

5.1 Complete Pipeline



5.2 Solid-Waste Algorithm

7. Load the existing YOLO-format solid-waste dataset.
8. Merge or remove classes only when required by the project's actual target categories.
9. Use YOLO11s pretrained weights and fine-tune on the waste dataset.
10. Evaluate precision, recall, F1, mAP50 and mAP50–95.
11. Run the trained model on images/video frames.
12. Return waste class, bounding box and confidence.
13. Count detected objects inside the camera's region of interest.
14. Optionally apply ByteTrack to prevent the same object from being counted repeatedly in video.

Why detection? A bottle, bag or can is an individual object. Bounding boxes provide its location and allow the system to count objects and estimate the amount of visible solid waste.

5.3 Liquid-Waste Algorithm

15. Download the existing spill-segmentation dataset and inspect its class definitions.
16. Convert/export masks into the format required by the chosen YOLO11-Seg training pipeline.
17. Train YOLO11s-Seg using pretrained weights.
18. Evaluate mask IoU/mIoU, Dice/F1, precision and recall.
19. Run segmentation on a new image/frame.
20. Calculate the number of pixels inside the predicted liquid mask.
21. Divide liquid-mask pixels by the monitored region-of-interest pixels to obtain liquid coverage percentage.

Liquid Coverage (%) = Liquid-mask pixels ÷ ROI pixels × 100

Why segmentation? A liquid spill has an irregular shape. A bounding box may contain a large amount of clean floor, while a mask provides a closer estimate of the actual affected region.

5.4 Data Preparation

Step	Action	Purpose
1	Download public dataset	Avoid creating a training database from scratch.
2	Inspect labels/classes	Remove invalid labels and identify rare categories.
3	Map classes	Make the dataset match the project's final taxonomy.
4	Resize/normalize	Standardize model input.
5	Augment training images	Improve robustness to viewpoint, brightness and scale.
6	Split by scene/source where possible	Reduce train-test leakage, especially for video frames.
7	Train → validate → test	Select model using validation; report final

		result only on held-out test data.

6. Cleanliness and Severity Algorithm

The AI models are the perception layer. A simple rule-based severity engine converts their outputs into an understandable cleanliness status. This makes the project a monitoring system rather than only an object detector.

Input	Measurement	Example contribution
Solid waste	Unique object count / density	More objects → higher solid severity
Liquid waste	Liquid coverage %	Larger affected area → higher liquid severity
Confidence	Average/individual confidence	Low-confidence detections can be ignored or reviewed
Persistence	Time detected in video	Longer persistence → higher priority
Zone	Camera/ROI importance	Critical zones can receive higher operational priority

Prototype formula: Cleanliness Score = 100 − (w_s × Solid Severity + w_l × Liquid Severity). The weights and thresholds are project parameters, not universal cleanliness standards; they should be validated against human assessment during testing.

Score	Status	Suggested response
80–100	Clean	Routine monitoring
60–79	Acceptable	Continue monitoring
40–59	Moderate	Cleaning attention
20–39	Dirty	High-priority cleaning
0–19	Critical	Immediate review

7. Implementation Plan

Phase	Work to complete	Expected output
Phase 1	Download and inspect both public datasets; define final classes.	Clean dataset folders + class mapping
Phase 2	Train YOLO11s on solid-waste dataset.	Solid detection model
Phase 3	Train YOLO11s-Seg on spill dataset.	Liquid segmentation model
Phase 4	Build one inference program that runs both models.	Dual AI inference
Phase 5	Add object count and liquid-area calculation.	Quantified waste information
Phase 6	Add severity/cleanliness scoring and thresholds.	Cleanliness status
Phase 7	Add video input and ByteTrack.	Near-real-time monitoring
Phase 8	Build dashboard and alert display.	Working demonstration system
Phase 9	Evaluate, perform error analysis and document results.	Final model + report

8. Evaluation Plan

Component	Metrics	What it tells us
Solid YOLO11	Precision, Recall, F1, mAP50, mAP50–95	Detection and localization quality
Liquid YOLO11-Seg	IoU/mIoU, Dice/F1, Precision, Recall	Quality of predicted liquid masks
Tracking	ID consistency / tracking metrics where applicable	Whether objects are counted persistently
System	Alert precision, missed-alert rate, latency/FPS	Practical usefulness of the complete system

Testing should include difficult cases: small litter, occlusion, low light, shadows, clutter, reflective surfaces, transparent liquid, wet floors and confusing background textures. A visual error-analysis table should be included in the final report.

9. 3rd-Year Enhancements

- Zone-wise monitoring: assign each camera/frame to a monitored area and compare cleanliness across zones.
- Persistence-aware alerts: trigger a stronger alert when waste remains detected for a configured duration.
- Before/after cleaning: compare cleanliness scores before and after a cleaning event.
- Historical analytics: show daily/weekly counts and liquid-coverage trends using stored experiment outputs.
- Human verification: allow incorrect AI detections to be marked for future retraining.
- Risk prediction: after enough historical observations are collected, train a small model to predict high-risk time/zone combinations.

These enhancements should be added after the two core models work reliably. The priority is to finish a complete, measurable dual system rather than adding many partially implemented features.

10. Dataset Access

Recommended download sources:

Solid — Kaggle: Garbage Detection – 6 Waste Categories
https://www.kaggle.com/datasets/viswaprakash1990/garbage-detection
10,464 YOLO-labeled images; 6 classes; train/validation/test split already supplied.

Liquid — Roboflow: Spill Instance Segmentation Dataset
https://universe.roboflow.com/shah-alam/spill-r2tsg
4,500 images; instance segmentation; water/trash/spill classes; CC BY 4.0.

Liquid — LQDS paper/dataset reference
https://arxiv.org/abs/2601.00940
5,000 real-world liquid images with pixel-level masks across 14 liquid classes.

11. Final Recommended Configuration

Item	Final recommendation
Solid dataset	Kaggle Garbage Detection – 6 Waste Categories (10,464 images)
Solid model	YOLO11s → compare YOLO11m only if needed
Liquid dataset	Roboflow Spill Instance Segmentation (4,500 images)
Liquid model	YOLO11s-Seg → compare U-Net if time permits
Video	ByteTrack as Phase-2 enhancement
Scoring	Transparent rule/weighted cleanliness score
Database development	Not required for Milestone 2; use existing public datasets
Dashboard	Streamlit for the first prototype
Main project contribution	Solid detection + liquid segmentation + quantitative cleanliness assessment + actionable alerts

12. Conclusion

The proposed Milestone 2 design uses two task-specific AI models and existing public datasets. YOLO11s is selected for solid-waste object detection because the system needs object class, location and count. YOLO11s-Seg is selected for liquid waste because pixel-level masks allow the affected spill area to be estimated. The project does not require creating a new database for model training. Instead, the team can download the recommended datasets, prepare their labels, train the models, and integrate their outputs into a cleanliness and alerting pipeline. This provides a realistic and sufficiently advanced 3rd-year project scope while leaving room for tracking, analytics and prediction enhancements.

Important: Dataset licenses and download availability should be checked on the source page immediately before final submission. Do not combine train/test images from the same video or near-duplicate source across splits.